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ORIGINAL PAPER

Carotid artery mechanical properties and stresses quantified using in vivo data from normotensive and hypertensive humans

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Abstract The goal of this study was to model the in vivo non-linear mechanical behavior of human common carotid arteries (CCAs) and then to compare wall stresses and associated contributions of micro-constituents in normotensive (NT) and treated hypertensive (HT) subjects. We used an established theoretical model of 3D arterial mechanics that assumes a hyperelastic, anisotropic, active–passive, and residually stressed wall. In vivo data were obtained non-invasively from CCAs in 16 NT (21–64 years old) and 25 treated HT (44–69 years old) subjects. The associated quasi-static boundary value problem was solved semi-analytically over a cardiac cycle while accounting for surrounding perivascular tissue. Best-fit values of model parameters, including those describing contributions by intramural elastin, fibrillar collagen, and vascular smooth muscle, were estimated by a non-linear least-squares method. The model (1) captured temporal changes in intraluminal pressure, (2) estimated wall stress fields that appeared to reflect the presence or absence of age and disease, and (3) suggested changes in mechanical characteristics of wall micro-constituents despite medical treatment of hypertension. For example, age was positively correlated with residual stresses and altered fibrillar collagen in NT subjects, which indirectly validated the modeling, and HT subjects had higher levels of stresses, increased smooth muscle tone, and a stiffer elastin-dominated matrix despite treatment. These results are consistent with prior reports on effects of age and hypertension, but provide increased insight into evolving contributions of cell and matrix mechanics to arterial behavior in vivo.

Keywords Anisotropic hyperelasticity · Microstructural models · Aging · Hypertension

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1 Introduction

Cardiovascular diseases remain the number one cause of death and are appropriately the subject of intense research. With regard to arteries, one of the primary needs is to improve our understanding of biological mechanisms involved in aging as well as in diseases such as hypertension, atherosclerosis, and the formation of aneurysms. Because of the importance of mechanobiology in such diseases, physical modeling of mechanical properties and associated distributions of stress within large arteries promises to provide a better understanding of disease progression and hence suggest improved therapeutic approaches.

Numerous models have been developed to describe the mechanical behavior of arteries. Recently, however, classical phenomenological models (Fung 1993; Humphrey 2002) have given way to models that are motivated by the underlying composition and micro-structure of the wall (Baek et al. 2007; Gasser and Holzapfel 2002; Gasser et al. 2006; Holzapfel et al. 2000, 2005; Masson et al. 2008;

Zulliger et al. 2004). Such structurally motivated models promise to provide physically more meaningful parameter values, yet they must be based on appropriate experimental data. Whereas most prior mechanical studies have relied on in vitro tests, which are fundamental to formulating and testing constitutive relations, there is a pressing need for in vivo data to refine these models and render them useful clinically. In vivo behavior can differ from that in vitro because in vitro specimens are deprived of their natural biological environment, including hormonal stimuli and perivascular tethering. Although considerable in vivo data are available, they tend to be incomplete because the data collection was not motivated by appropriate non-linear modeling.

Notwithstanding inherent shortcomings of in vivo mechanical data, such as the lack of information on axial stretch and perivascular tethering, recent studies demonstrate the feasibility of characterizing complex mechanical properties and stress fields from in vivo clinical data (Masson et al. 2008; Schulze-Bauer and Holzapfel 2003; Stalhand 2009; Stalhand et al. 2004). Including micro-structural complexity in such models enables one to estimate contributions of individual constituents (e.g., roles of elastin as it fragments or degrades, roles of fibrillar collagen as it turns over, and roles of smooth muscle cells as they proliferate, experience apoptosis, or change phenotype) to measurable global responses, which in turn may suggest alternate treatment strategies. That is, changes in cellular and matrix composition and function play important roles in the mechanical response of arteries and associated disease progression (Humphrey 2002). Although identification of parameter values in complex models can be complicated by the existence of multiple local minima (Ferruzzi et al. 2010), results from in vitro tests provide important bounds for the estimation. Moreover, parametric studies can increase confidence in estimated values (Eddhahak-Ouni et al. 2009). Fundamental to comprehensive estimation, however, is inclusion of key features of arterial behavior, including the existence of residual stresses, presence of tone by the smooth muscle, effects of perivascular tissues, and time-varying loading over a cardiac cycle.

We previously demonstrated the feasibility of estimating material, structural, and functional parameters in a microstructurally motivated model of the carotid wall directly from non-invasive clinical data (Masson et al. 2008). The goals of this paper were to build on this success and to quantify differences in both mechanical properties and wall stresses between normotensive (NT) and treated hypertensive (HT) subjects. The proposed 3D model for common carotid arteries (CCAs) captures known hyperelastic, anisotropic, active-passive, and residually stressed mechanical characteristics.

2 Methods

2.1 Clinical human data

2.1.1 Study populations

The overall study group consisted of 16 NT subjects (8 women, 8 men) from 21 to 64 years of age, and 25 treated HT patients (12 women, 13 men) from 44 to 69 years of age; each subject signed an informed consent prior to investigation. For the purpose of comparison, however, a subgroup consisting of 11 NT subjects from 39 to 64 years of age (7 women, 4 men) was selected from the 16 NT subjects for age-matching with the HT subjects. These NT subjects did not have any discernible disease or risk factors (e.g., no atherosclerosis, hypertension, or renal failure). The HT patients had essential, uncomplicated hypertension and were treated and well controlled for their high blood pressure (Table 1).

2.1.2 In vivo measurements

Routine clinical measurements were performed following 15 min of recumbent rest in a room dedicated to ultrasound examination. Specifically, both medial diameter (i.e., diameter at the medial-adventitial border) and intimal-medial thickness (IMT) were measured in the CCA using a high-resolution echotracking device (Art.Lab, Esaote) consisting of a 128 radio-frequency line multiarray applied over a 4-cm arterial segment (Beaussier et al. 2008; Paini et al. 2007). Geometric data were recorded for 6 s at 30 Hz in B-mode with a spatial resolution $\sim$21 μm (Meinders et al. 2001). Subsequently (i.e., 5 $\pm$ 2 min after echotracking recording), blood pressure was measured in the same CCA using an applanation tonometer incorporating a high-fidelity Millar strain gauge transducer (SphygmoCor, AtCor Medical). This pressure was recorded for 9 s at 128 Hz with a resolution $\sim$2.5 mmHg (Boutouyrie et al. 1999; Kelly et al. 1989). Subjects were asked to hold their breath for 10 s during the acquisitions of arterial dimensions and blood pressures.

2.1.3 Data processing

Data were first processed to yield a representative "mean cardiac cycle" for both inner diameter and intraluminal pressure based on their respective consecutively measured cycles; using mean data reduces undesirable disturbances (like Mayer waves) and improves parameter identification (Stalhand 2009). Moreover, because geometry (medial diameter and IMT) and applied load (blood pressure) were measured over successive, not simultaneous, cardiac

Table 1 Demographic and hemodynamic characteristics for normotensive (NT) subjects ($n = 16$) and treated hypertensive (HT) patients ($n = 25$)

Parameters	NT ($n = 16$)	HT ($n = 25$)	Matched NT ($n = 11$)	p (HT versus matched NT)
Age (years)	47.3 ± 15.0	58.8 ± 6.5	55.8 ± 8.2	0.48
Sex (M/F)	8/8	13/12	4/7	0.39
BMI (kg/m^2)	24.3 ± 4.4	25.8 ± 3.0	24.2 ±3.5	0.24
SBP brachial (mmHg)	115.9 ± 13.1	127.3 ± 12.4	116.3 ± 12.8	0.03
DBP brachial (mmHg)	64.2 ± 9.2	72.6 ± 9.2	67.2± 7.6	0.07
MBP brachial (mmHg)	81.1 ± 8.9	90.9 ± 9.9	83.5± 8.3	0.07
Heart rate (bpm)	69.4 ± 8.1	67.6 ± 12.1	67.5 ±8.1	0.81
Smoking status (current/past/never)	1/14/1	7/16/2	1/9/1	0.45
Hypercholesterolemia (yes/no)	5/11	10/15	5/6	0.76
Diabetes mellitus (yes/no)	1/15	2/23	1/10	0.91
Antihypertensive treatment (yes/no)	0/16	25/0	0/11	–
ACEI or ARB (yes/no)	–	23/2	–	–
Calcium channel blocker (yes/no)	–	13/12	–	–
Diuretic (yes/no)	–	18/7	–	–
β-blocker (yes/no)	–	7/18	–	–

Statistical results are from comparisons between the HT patients and the subgroup of age-matched NT subjects ($n = 11$, selected from the 16 NT subjects)
Quantitative results expressed as mean ± standard deviation. Significance level: $p < 0.05$ (Wilcoxon rank-sum test for quantitative variables, and χ^2 test for categorical variables)
M Male, *F* female, *BMI* body mass index, *BP* blood pressure, *S* systolic, *D* diastolic, *M* mean, *ACEI* angiotensin-converting enzyme inhibitor, *ARB* angiotensin II receptor blocker

cycles, these mean data were aligned temporally prior to parameter estimation. Toward this end, note that prior high-resolution recordings with a simultaneous electrocardiogram revealed that tracings of inner diameter and pressure correlate perfectly during the systolic (or, distension) phase (Masson et al. 2008). Recordings of inner diameter and pressure were thus segmented into individual cardiac cycles, each of which was normalized over time from 0 to 1 and re-sampled to obtain mean cycles. The mean cycles of inner diameter and pressure were then aligned temporally based on the distension phase. These mean, normalized cycles of inner and medial diameters and intraluminal pressure thus provided consistent input experimental data for theoretical modeling of the mechanical behavior and enabled direct comparisons across subjects.

2.2 Mechanical modeling

From a macroscopic point of view, arterial tissue can be considered as a continua subjected to finite strains (Humphrey 2002). Hence, the mechanical behavior was quantified using an inverse approach to identify continuum mechanical properties of the wall based on the in vivo data, as proposed previously (Masson et al. 2008). The main steps are presented below.

2.2.1 Kinematics

We considered measured segments of the CCAs to be thick-walled circular cylinders. The existence of residual stresses was accounted for by assuming an open sector (resulting from a fictitious stress-relieving radial cut; Chuong and Fung 1986) as the reference configuration. Residual stresses result from, among other causes, non-uniform growth and remodeling during development as well as possibly during adaptations to altered hemodynamics. Relative to a cylindrical coordinate system defined by an origin and basis $(\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_z)$, the total deformation experienced by a CCA was described via two successive motions (Fig. 1). The first maps material particles from the assumed stress-free reference configuration (Ω_0) to an unloaded in vitro configuration (Ω_1), which accounts for residual stresses via the parameter Θ_0 (i.e., opening angle) and the axial stretch ratio Λ. The second maps material particles from the unloaded configuration (Ω_1) to in vivo configurations (Ω_2) that vary with time t over the cardiac cycle, with λ accounting for the in vivo load-induced axial stretch (assumed to be constant over the cardiac cycle; Humphrey et al. 2009; van Loon 1977). The motion is thus given by

$$r = r\,(R, t)\,, \quad \theta = \frac{\pi}{\Theta_0}\Theta, \quad z = \lambda\Lambda Z \tag{1}$$

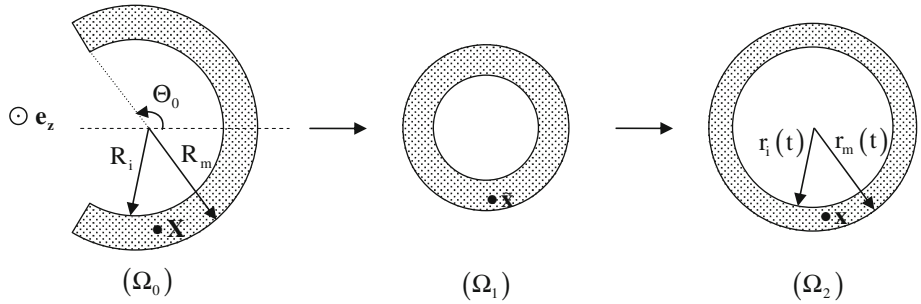


Fig. 1 Kinematics of the arterial wall relative to a nearly stress-free reference configuration (Ω_0), with mappings to the unloaded excised configuration (Ω_1) and subsequently to in vivo time-dependent configurations (Ω_2), with cylindrical coordinates of $\mathbf{X}(R, \Theta, Z)$, $\tilde{\mathbf{x}}(\rho, \phi, \xi)$, and $\mathbf{x}(r(t), \theta, z)$, respectively

where R_i and $r_i(t)$ denote inner radii of the CCA in the reference and in vivo configurations, respectively. Similarly, R_m and $r_m(t)$ denote radii at the media–adventitia interface (also called medial radii, not to be confused with mid-wall radii). Consequently, the deformation gradient tensor $\mathbf{F}$ can be represented in terms of principal stretches ($\lambda_r, \lambda_\theta, \lambda_z$ in radial, circumferential, and axial directions, respectively) as

$$\mathbf{F} = \begin{bmatrix} \frac{\partial r}{\partial R} & 0 & 0 \\ 0 & \frac{\pi r}{\Theta_0 R} & 0 \\ 0 & 0 & \lambda\Lambda \end{bmatrix} = \begin{bmatrix} \lambda_r & 0 & 0 \\ 0 & \lambda_\theta & 0 \\ 0 & 0 & \lambda_z \end{bmatrix}. \tag{2}$$

The left and right Cauchy-Green tensors are defined as $\mathbf{B} = \mathbf{F}\mathbf{F}^{\mathrm{T}}$ and $\mathbf{C} = \mathbf{F}^{\mathrm{T}}\mathbf{F}$ where $\mathbf{F}^{\mathrm{T}}$ is the transpose of $\mathbf{F}$. Furthermore, the reference radius can be found by using the incompressibility condition. If $J = \det\mathbf{F} = \lambda_r\lambda_\theta\lambda_z = 1$, then

$$R = \left(R_m^2 - \frac{\pi\lambda\Lambda}{\Theta_0}\left(r_m^2 - r^2\right)\right)^{1/2}. \tag{3}$$

Note that the residual stress related axial stretch Λ is typically near unity (Chuong and Fung 1983; Humphrey 2002), thus we let $\Lambda = 1$.

2.2.2 Anisotropic constitutive relation

The arterial wall consists primarily of amorphous elastic fibers, oriented fibrillar collagen, and oriented smooth muscle capable of contracting. Assuming that the passive portion of the behavior is nearly elastic (hyperelastic), the strain energy function W is written in terms of an isotropic part representing the elastin-dominated matrix and an anisotropic representing $k = 1, 2, 3, 4$ primary families of collagen fibers (Baek et al. 2007; Masson et al. 2008), which is an extension of an earlier two-fiber family model (Holzapfel et al. 2000):

$$W = \frac{c}{2}(I_1 - 3) + \sum_{k=1}^{4} \frac{c_{1(k)}}{4c_{2(k)}}\left\{\exp\left[c_{2(k)}\left(I_{4(k)} - 1\right)^2\right] - 1\right\}, \tag{4}$$

where the strain invariants I_j ($j = 1, 4$) are given by

$$I_1 = tr\mathbf{C}, \quad \mathbf{I}_{4(k)} = \mathbf{M}_{\alpha_k}\left(\mathbf{C}\mathbf{M}_{\alpha_k}\right) \quad (k = 1, 2, 3, 4). \tag{5}$$

Here, c, $c_{1(k)}$, $c_{2(k)}$ are material parameters and $\mathbf{M}_{\alpha_k}$ are unit vectors that denote the collagen fiber orientations in the reference configuration, which depend on angles α_k defined between the direction of the kth family of fibers and the axial direction of the artery (Fig. 2). Note that these fiber directions $\mathbf{M}_{\alpha_k}$ are related to directions in the in vivo deformed state as $\mathbf{a}_{\alpha_k} = \mathbf{F}\mathbf{M}_{\alpha_k}$, which includes the stretching of the fiber due to deformation.

Based on (4) and the contribution by the smooth muscle tone, the Cauchy stress tensor $\boldsymbol{\sigma}$ for anisotropic incompressible CCA tissue takes the form

$$\boldsymbol{\sigma} = -p\mathbf{1} + \boldsymbol{\sigma}^{\mathbf{p}} + \boldsymbol{\sigma}^{\mathbf{a}}, \tag{6}$$

where p is a Lagrange multiplier that enforces incompressibility during transient motions and $\mathbf{1}$ is the identity tensor. The tensors $\boldsymbol{\sigma}^{\mathbf{p}}$ and $\boldsymbol{\sigma}^{\mathbf{a}}$ represent the passive and active stress contributions (Humphrey 2002) given by

$$\begin{aligned} \boldsymbol{\sigma}^{\mathbf{p}} &= 2W_1\mathbf{B} + 2W_4\mathbf{F}\mathbf{M}_{\alpha_k} \otimes \mathbf{M}_{\alpha_k}\mathbf{F}^{\mathrm{T}}, \\ \boldsymbol{\sigma}^{\mathbf{a}} &= T_m\lambda_\theta\left[1 - \left(\frac{\lambda_m - \lambda_\theta}{\lambda_m - \lambda_0}\right)^2\right]\mathbf{e}_\theta \otimes \mathbf{e}_\theta, \end{aligned} \tag{7}$$

with $W_j = \partial W/\partial I_j$ ($j = 1, 4$). Furthermore, we assume that the smooth muscle is oriented primarily in the $\mathbf{e}_\theta$ direction (cf. Walmsley et al. 1983). The parameter T_m represents the level of activation, λ_m and λ_0 are, respectively, the

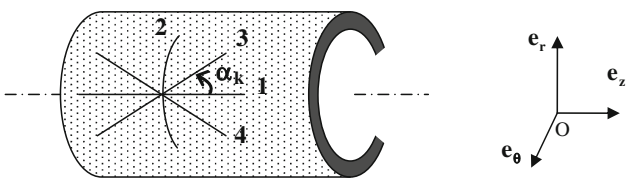


Fig. 2 Schema of the four collagen-fiber families, with the diagonal fibers characterized by angle α relative to the axial direction in a cylindrical arterial segment

stretch at which contraction is a maximum and the stretch at which active force generation ceases (Humphrey and Na 2002; Rachev and Hayashi 1999). From (6) and (7), with $\mathbf{M}_{\alpha_k} = [0, \sin\alpha_k, \cos\alpha_k]$, non-zero components of the Cauchy stress tensor are

$$\begin{aligned}
\sigma_{rr}(r,t) &= -p(r,t) + c\lambda_r(r,t)^2, \\
\sigma_{\theta\theta}(r,t) &= -p(r,t) + c\lambda_\theta(r,t)^2 \\
&\quad + \lambda_\theta(r,t)^2 \sum_{k=1}^{4} \left\{ c_{1(k)} \left(\lambda_k(r,t)^2 - 1 \right) \right. \\
&\quad \left. \times \exp\left[c_{2(k)} \left(\lambda_k(r,t)^2 - 1 \right)^2 \right] \sin^2\alpha_k \right\} \\
&\quad + T_m \lambda_\theta(r,t) \left[1 - \left(\frac{\lambda_m - \lambda_\theta(r,t)}{\lambda_m - \lambda_0} \right)^2 \right], \\
\sigma_{zz}(r,t) &= -p(r,t) + c\lambda_z(r,t)^2 \\
&\quad + \lambda_z(r,t)^2 \sum_{k=1}^{4} c_{1(k)} \left(\lambda_k(r,t)^2 - 1 \right) \\
&\quad \times \exp\left[c_{2(k)} \left(\lambda_k(r,t)^2 - 1 \right)^2 \right] \cos^2\alpha_k, \\
\sigma_{\theta z}(r,t) &= \lambda_\theta(r,t)\lambda_z(r,t) \sum_{k=1}^{4} c_{1(k)} \left(\lambda_k(r,t)^2 - 1 \right) \\
&\quad \times \exp\left[c_{2(k)} \left(\lambda_k(r,t)^2 - 1 \right)^2 \right] \sin\alpha_k \cos\alpha_k,
\end{aligned} \tag{8}$$

where $\lambda_k = \left(\lambda_\theta^2 \sin^2\alpha_k + \lambda_z^2 \cos^2\alpha_k \right)^{1/2}$ is the stretch of the kth fiber family.

Moreover, we assumed that one collagen fiber family is oriented axially ($\alpha_1 = 0°$) and another circumferentially ($\alpha_2 = 90°$). Hence, only one fiber angle is unknown. Assuming that the two other (diagonal) families of fibers are oriented symmetrically about the axial direction (i.e., $\alpha_3 = \alpha$ and $\alpha_4 = -\alpha$), we see that the shear stresses in (8) disappear, hence yielding a principal state of stress. Furthermore, we let the diagonal and axial collagen fibers have similar behaviors (i.e., $c_{1(1)} = c_{1(3)} = c_{1(4)} = c_1$ and $c_{2(1)} = c_{2(3)} = c_{2(4)} = c_2$), whereas the circumferential fibers may differ ($c_{1(2)} = c_{1\text{circ}}$ and $c_{2(2)} = c_{2\text{circ}}$). This assumption is based on possible contributions of passive smooth muscle to the overall circumferential mechanical behavior.

2.2.3 Boundary-value problem

In the absence of body forces, the equation of motion reduces to

$$\frac{\partial \sigma_{rr}}{\partial r} + \frac{\sigma_{rr} - \sigma_{\theta\theta}}{r} = \rho a_r, \tag{9}$$

where ρ and a_r are, respectively, the mass density of the wall and the radial acceleration in the current configuration. We assume, however, that the effect of the inertial term is negligible compared to the distending pressure (estimated at ~0.1% in Masson et al. 2008), consequently the elastodynamic problem can be studied quasi-statically (Humphrey and Na 2002). Although the arterial wall is layered, the in vivo clinical data on the CCA do not delineate the adventitia. Consequently, we focused on the mechanical behavior of the intimal-medial layer and assumed that the pressure-like contribution of the adventitial layer on the intima-media (i.e., radial traction) can be combined with that due to perivascular tissue and be described by a single term $P_a(t)$. The boundary conditions then depend on the intraluminal pressure $P_i(t)$ at the inner wall (due to the blood) and the perivascular-like pressure $P_a(t)$ at the medial wall (due to the perivascular tissue and action of the adventitia on the media):

$$\sigma_{rr}(r_i,t) = -P_i(t), \quad \sigma_{rr}(r_m,t) = -P_a(t). \tag{10}$$

Thus, by integrating (9) over radius between the measured inner $r_i(t)$ and medial $r_m(t)$ radii, we obtain the computed intraluminal pressure $P_i(t)$ acting on the intimal-medial tissue:

$$P_i(t) = P_a(t) + \int_{r_i(t)}^{r_m(t)} \frac{\sigma_{\theta\theta}(r,t) - \sigma_{rr}(r,t)}{r} dr. \tag{11}$$

The perivascular-like reactive radial stress is modeled using a lumped parameter exponential form that varies over the cardiac cycle (Humphrey and Na 2002):

$$P_a(t) = a \exp\left(b \frac{r_m(t)}{r_m(t_d)} \right), \tag{12}$$

where a, b and t_d are material parameters and the diastolic time over the cardiac cycle, respectively. In summary to compute the intraluminal pressure (11), the boundary-value problem given by (9) and (10) is solved semi-analytically.

2.2.4 Identification of model parameters

Using an inverse approach, we then estimated the unknown geometric, material, and loading parameters using the aforementioned non-invasively measured in vivo data on luminal and medial diameters and intraluminal pressure (cf. Masson et al. 2008). That is, (3), (4), (7), (12) and related assumptions yield 14 unknown parameters to be determined: residual stress related (R_m, Θ_0, λ), passive wall properties ($c, c_1, c_2, c_{1\text{circ}}, c_{2\text{circ}}, \alpha$), active wall properties ($T_m, \lambda_0, \lambda_m$), and perivascular-like effects on the media (a, b). Fortunately, many of the parameters are well bounded physically, which aids greatly in the identification. Specifically, best-fit values of these parameters were determined using a non-linear least-squares regression (function "lsqnonlin" in MATLAB) to minimize the difference between

computed and measured intraluminal pressures over a mean cardiac cycle, that is, by minimizing the objective function:

$$e = \sum_{j=1}^{N} \left[\left(P_i^{\text{th}}(\mathbf{u}) - P_i^{\text{exp}} \right)_j^2 \right], \tag{13}$$

where N is the number of data points of experimentally measured intraluminal pressure P_i^{exp}, P_i^{th} is the theoretically computed inner pressure from (11), and $\mathbf{u}$ is the vector of 14 parameters to be optimized. To ensure physical meaning of the unknown model parameters, initial guesses as well as lower and upper bounds were specified based on prior work using in vitro data (Delfino et al. 1997; Fridez et al. 2001; Hariton et al. 2007; Humphrey and Na 2002; Rachev and Hayashi 1999). Furthermore, optimal solutions were subsequently perturbed and parameter identifications were reperformed to check that the solutions were not saddle points and could be considered as good solutions. Finally, the quality of fit was estimated through the root mean square of the fitting error as

$$\text{RMSE} = \sqrt{\frac{e}{N}}. \tag{14}$$

2.3 Statistics

We performed correlation tests between model parameters and age in NT subjects by computing Spearman-rank correlation matrices; we chose non-parametric statistical tests to examine relationships between variables. We also performed a Wilcoxon rank sum test to compare model parameters for the NT and treated HT populations, and the categorical variables were compared using a χ^2 test. The statistical tests were performed using NCSS 2001 software (Hintze J., USA) and a p-value less than 0.05 was considered significant.

3 Results

3.1 Evolving properties and stresses with age in NT subjects

Demographic and hemodynamic information is given in Table 1 for the normotensive (NT) subjects. Associated theoretical predictions of intraluminal pressures corresponded well with in vivo measurements (Fig. 3a). In particular, systolic (distension phase) portions of the data agreed very well while diastolic (relaxation phase) portions were less in accordance but still close with a mean difference between theoretical and experimental results of ∼0.4 kPa (i.e. ∼3 mmHg). Best-fit values of the model parameters are in Table 2.

Study of possible changes in model parameters as a function of age for all 16 NT subjects revealed strong correlations for particular parameters (Fig. 4); associated Spearman correlation coefficients and significance levels are in Table 3 for all 14 parameters. In particular, the opening angle Θ_0 (cf. Fig. 1) decreased significantly with age ($r = -0.67$, $p = 0.005$), noting that smaller values imply a larger opening of the reference sector and hence larger residual stresses. The increase in R_m (i.e., medial radius in the reference configuration, cf. Fig. 1) with age ($r = 0.50$, $p = 0.05$) was consistent with this change in opening angle. The parameters $c_{1\text{circ}}$, $c_{2\text{circ}}$, and α (which respectively model circumferential stiffness due to collagen fibers directly and indirectly) each increased with age ($r = 0.64$, $p = 0.007$, $r = 0.66$, $p = 0.005$, and $r = 0.62$, $p = 0.01$). A concomitant increase in $c_{1\text{circ}}$ and $c_{2\text{circ}}$ may suggest an increasein collagen fiber stiffness (e.g., cross-linking) or an increase in the fraction of collagen oriented in

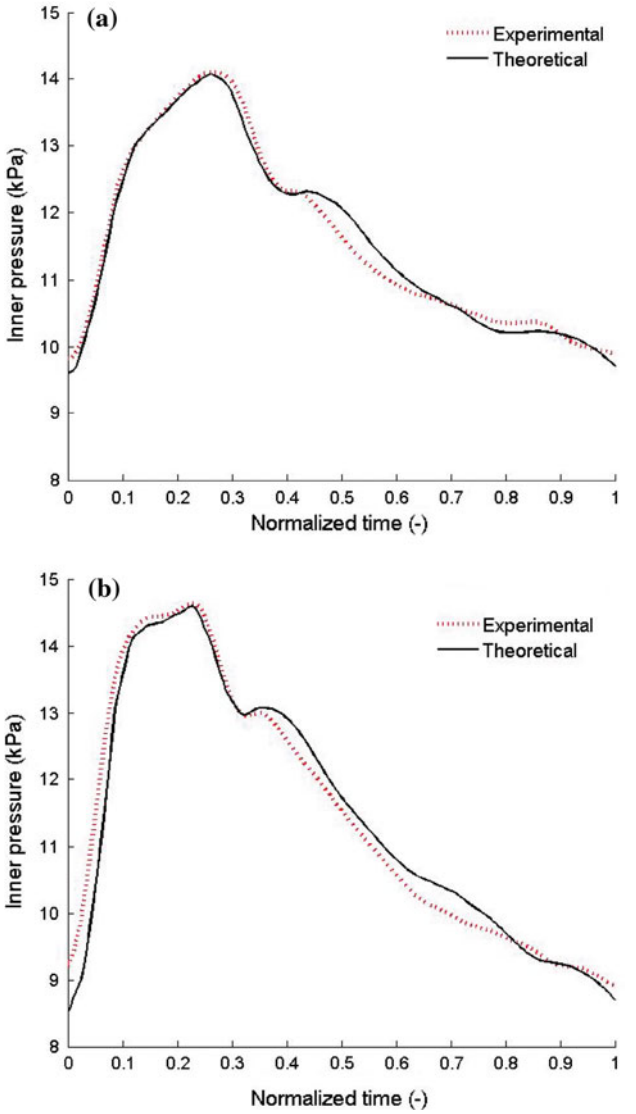


Fig. 3 Illustrative results of experimentally measured and theoretically computed intraluminal pressures for **a** a normotensive subject (NT #1, 62- year-old female) and **b** a treated hypertensive patient (HT #23, 51-year-old male)

Table 2 Best-fit values of the model parameters for the 16 normotensive subjects

Subjects NT #	Age (years)	Gender	R_m (mm)	Θ_0(°)	λ (−)	c (kPa)	c_1 (kPa)	c_2 (−)	$c_{1\text{circ}}$ (kPa)	$c_{2\text{circ}}$ (−)	α(°)	T_m (kPa)	λ_0 (−)	λ_m (−)	a (kPa)	b (−)	RMSE (kPa)
1	62	F	4.261	128.7	1.11	29.82	9.44	14.14	16.13	15.11	65.74	39.73	0.96	1.70	0.122	3.52	0.174
2	60	M	4.691	128.9	1.08	20.42	8.47	13.10	14.16	15.12	44.46	41.84	0.68	1.92	0.126	3.40	0.288
3	31	M	4.305	132.2	1.09	27.23	10.34	13.10	7.01	12.82	48.62	43.37	0.82	1.65	0.139	3.57	0.131
4	31	M	4.672	133.9	1.10	23.59	6.58	11.77	6.38	10.69	32.04	39.81	0.68	1.59	0.118	3.53	0.146
5	34	M	4.203	130.1	1.10	35.68	13.62	13.18	13.38	13.17	50.72	50.07	0.86	1.62	0.103	3.58	0.123
6	53	F	3.957	130.1	1.11	33.37	11.38	13.63	13.30	14.26	57.14	46.59	0.95	1.69	0.093	3.44	0.269
7	21	F	3.827	132.0	1.10	44.28	21.79	13.17	8.27	12.73	51.15	53.58	0.92	1.62	0.111	3.58	0.215
8	25	M	4.856	133.2	1.07	23.25	6.34	12.93	8.11	13.17	42.93	31.90	0.82	1.77	0.173	3.57	0.367
9	42	F	4.506	129.5	1.10	34.90	16.14	13.58	19.64	13.89	57.21	47.01	0.93	1.65	0.084	3.55	0.209
10	55	M	4.770	129.1	1.10	38.57	36.31	13.31	56.88	13.61	54.22	46.36	0.89	1.66	0.006	3.75	0.652
11	64	M	5.566	129.5	1.10	34.89	40.16	13.45	66.88	13.53	54.54	41.12	0.85	1.79	0.005	3.23	0.375
12	60	F	4.876	130.1	1.09	33.94	10.78	13.95	12.79	15.42	60.97	45.77	0.93	1.73	0.099	3.37	0.362
13	61	F	4.924	130.6	1.09	35.15	17.69	13.43	18.40	13.22	55.04	45.59	0.94	1.66	0.085	3.51	0.240
14	39	F	3.539	131.5	1.09	28.97	8.22	13.13	19.02	13.16	50.22	41.30	0.89	1.67	0.105	3.53	0.262
15	58	F	3.922	132.4	1.08	26.71	15.79	13.36	16.68	13.11	53.60	35.32	0.95	1.71	0.166	3.60	0.302
16	60	M	5.718	129.7	1.10	27.79	28.23	13.46	57.77	13.63	55.73	36.50	1.05	1.77	0.141	3.58	0.283
Mean	47	–	4.537	130.7	1.09	31.16	16.33	13.29	22.18	13.54	52.14	42.87	0.88	1.70	0.105	3.52	0.275
SD	15	–	0.599	1.6	0.01	6.28	10.36	0.52	19.57	1.13	7.81	5.59	0.10	0.08	0.047	0.12	0.129

SD Standard deviation, *F* female, *M* male

the circumferential direction; an increase in α (>45°), which defines the angle between diagonal fibers and the axial direction of the artery (cf. Fig. 2), may suggest a further increase in collagen fibers oriented toward the circumferential direction. Regardless, an overall increase in stiffness circumferentially is consistent with a decreased distensibility with age. The parameters c_1 and c_2 (related to collagen fibers in axial and diagonal directions) also increased with age, although only the change in c_2 was significant ($r = 0.66$, $p = 0.006$). This finding suggests that the stiffening may be biaxial. There was a positive correlation between λ_m and age ($r = 0.60$, $p = 0.01$), where λ_m is the stretch at which contraction of smooth muscle is maximum. This finding may be interpreted as a change in the contractile state of the vascular smooth muscle with age, which could result from intrinsic changes within the contractile apparatus, altered endothelial production of vasoactive molecules, or both. Finally, there was a significant decrease in the parameter b (related to mechanical effects of the adventitia and perivascular tissue on the intimal-medial layers), but only a trend toward a decrease for the associated parameter a (cf. Eq. 12).

Best-fit values of model parameters (cf. Table 2) allow one to compute wall stress transmurally (radial, circumferential, axial) over the cardiac cycle. Figure 5 shows computed circumferential and axial Cauchy stresses as a function of circumferential stretches for all 16 NT subjects. This population was arbitrarily divided into two groups for graphical presentation: the seven youngest subjects aged 21–42 years old (inverted triangles) and the nine oldest subjects aged 53–64 years old (circles). Mean values of stress, at each stretch, were computed for the two subgroups and included in Fig. 5. The overall results suggest a steeper slope (i.e. a higher wall circumferential stress gradient) and a reduced range of circumferential stretching for the older subjects compared to the younger subjects (Fig. 5a). This finding may be interpreted as increased arterial stiffening circumferentially and decreased distensibility with age. The overall mean behavior suggests that axial stresses are higher for the older subjects. In particular, mean circumferential and axial stresses were, respectively, 14 and 37% higher in older compared to younger subjects at the mean stretch of the outer wall.

3.2 Wall micro-constituents and stresses in HT patients

Demographic and hemodynamic information is given in Table 1 for the treated hypertensive (HT) patients as well as the subgroup of age-matched NT subjects. Although well controlled for high blood pressure (BP) through antihypertensive treatment, the HT subjects still had higher systolic, diastolic, and mean BP than did the NT subjects. Both groups were otherwise comparable with regard to body mass index, heart rate, and typical risk factors (smoking, hypercholesterolemia, and diabetes).

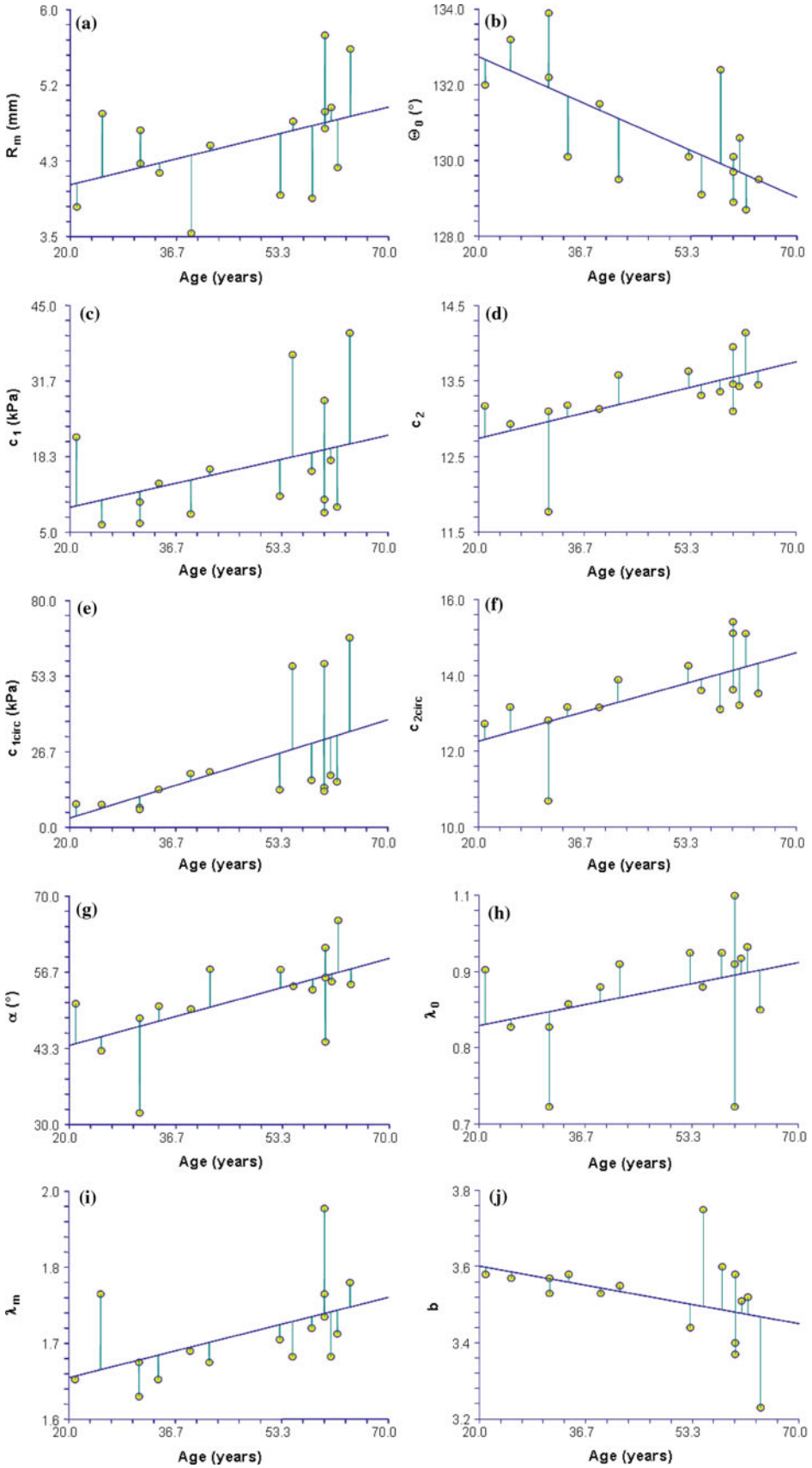


Fig. 4 Scatter plots for selected model parameters as a function of age for all normotensive subjects ($n = 16$). *Circles* represent parameter values for each subject whereas the *solid line* shows the regression, with corresponding residuals. **a** R_m, **b** Θ_0, **c** c_1, **d** c_2, **e** $c_{1\text{circ}}$, **f** $c_{2\text{circ}}$, **g** α, **h** λ_0, **i** λ_m, **j** b

Table 3 Correlations between model parameters and the age of all normotensive subjects ($n = 16$)

Model parameters	r	p
R_m (mm)	0.50	0.05
Θ_0 (°)	−0.67	0.005
λ (−)	0.13	0.64
c (kPa)	0.05	0.85
c_1 (kPa)	0.37	0.16
c_2 (−)	0.66	0.006
$c_{1\text{circ}}$ (kPa)	0.64	0.007
$c_{2\text{circ}}$ (−)	0.66	0.005
α (°)	0.62	0.01
T_m (kPa)	−0.24	0.37
λ_0 (−)	0.43	0.10
λ_m (−)	0.60	0.01
a (kPa)	−0.30	0.26
b (−)	−0.50	0.05

Spearman correlation coefficient r, and significance level $p < 0.05$

Best-fit values of model parameters are in Table 4, which again resulted in good predictions of intraluminal pressures relative to in vivo measurements (Fig. 3b). Model parameters were then compared between treated HT patients ($n = 25$) and the age-matched NT subjects ($n = 11$). Significant levels for the Wilcoxon rank sum test are in Table 5, which revealed significant differences between HT and NT groups for the parameter c (linked to the elastin-dominated matrix) and both T_m and λ_m (linked to contractile vascular smooth muscle behavior). These differences are revealed further by box plots (Fig. 6): the best-fit value of c was 20.5% higher for HT than NT ($p = 0.01$), T_m was 17.3% higher ($p = 0.003$), and λ_m was 2.9% lower ($p = 0.006$). In other words, the model suggested a left-ward shift, with augmentation, of the active smooth muscle behavior.

Transmural distributions of wall stress were again computed over the cardiac cycle using the best-fit values of the model parameters. Figure 7 compares results for circumferential and axial Cauchy stresses, as a function of circumferential stretch, for both the treated HT patients and age-matched NT subjects. Mean values of stress, at each stretch, were computed and compared for the two groups. Overall, stresses were higher within the intimal-medial portion of the wall of HT patients. Specifically, mean circumferential and axial stresses were, respectively, 19 and 30% higher in HT compared to NT subjects at the mean stretch of the outer wall. These increases did not reach statistical significance, however, for either the circumferential (61.9 ± 18.4 kPa in HT vs. 51.9 ± 18.0 kPa in age-matched NT; $p = 0.14$) or axial (33.0 ± 13.3 kPa in HT vs. 25.3 ± 13.1 kPa in age-matched NT; $p = 0.14$) stresses.

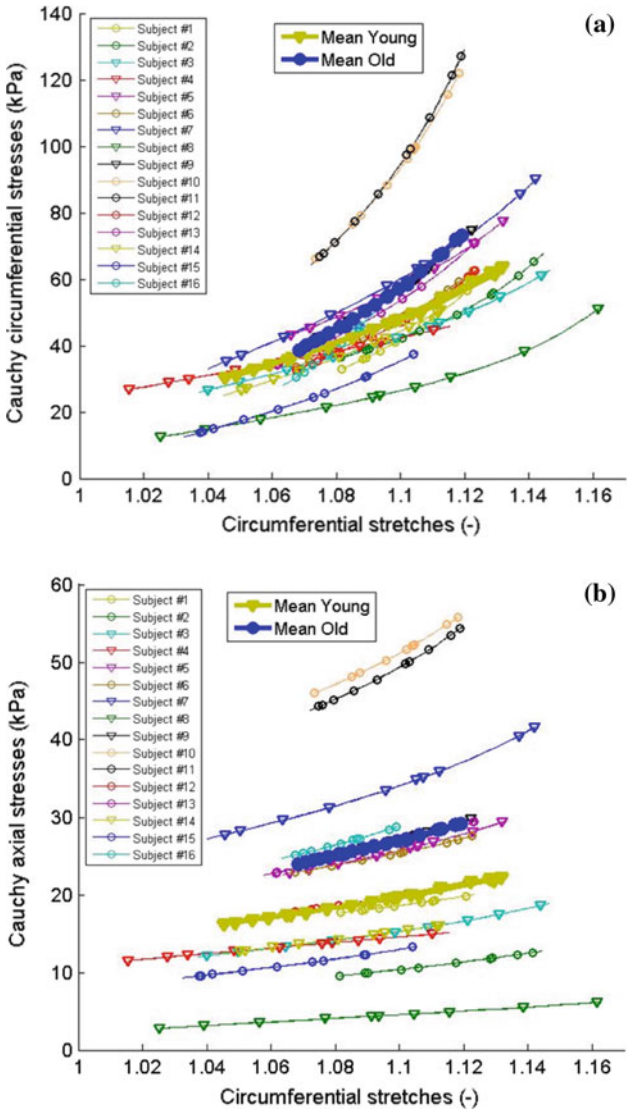


Fig. 5 **a** Circumferential and **b** axial Cauchy stresses as a function of circumferential stretches, expressed at the outer media r_m, for all 16 normotensive subjects. The seven youngest subjects (from 21 to 42 years old) are denoted by *inverted triangles* (∇) and the nine oldest ones (from 53 to 64 years old) by *circles* (o). The respective mean values are shown by *darker solid lines*: the "Young" group in ochre and the "Old" group in *blue*. Mean circumferential and axial stresses (expressed at the approximate mean stretch of the outer medial location) were, respectively, 14 and 37% higher in older compared to younger subjects. For display purposes, one point in twenty was plotted for clarity, although computations were performed on the full data sets

4 Discussion

This study employed a non-linear inverse approach to infer structural, mechanical, and functional characteristics of common carotid arteries from non-invasive in vivo measurements of wall dimensions and blood pressures in humans by minimizing differences between measured and computed pressures. The artery was modeled as a residually

Table 4 Best-fit values of the model parameters for the 25 treated hypertensive patients

Patients HT #	Age (years)	Gender	R_m (mm)	Θ_0 (°)	λ (−)	c (kPa)	c_1 (kPa)	c_2 (−)	$c_{1\text{circ}}$ (kPa)	$c_{2\text{circ}}$ (−)	α(°)	T_m (kPa)	λ_0 (−)	λ_m (−)	a (kPa)	b (−)	RMSE (kPa)
1	68	M	4.965	130.9	1.10	35.91	12.57	14.08	15.52	14.91	67.18	48.42	0.99	1.66	0.098	3.50	0.300
2	54	M	4.927	129.8	1.10	38.20	16.40	13.20	16.44	13.20	51.11	52.44	0.86	1.62	0.137	3.58	0.213
3	64	F	4.869	129.5	1.10	36.01	14.55	13.96	18.35	14.48	62.66	47.02	1.00	1.74	0.113	3.62	0.419
4	58	F	4.574	131.2	1.09	22.42	8.57	13.03	6.43	13.36	46.38	38.53	0.81	1.68	0.099	3.51	0.346
5	62	F	4.779	130.1	1.10	48.19	35.06	13.23	43.19	13.03	53.24	52.86	0.91	1.63	0.033	3.56	0.218
6	66	F	5.062	129.1	1.10	39.78	37.75	13.35	60.89	13.45	56.01	46.44	0.91	1.64	0.004	3.54	0.349
7	54	F	4.298	129.6	1.11	50.45	25.51	13.32	39.43	13.24	53.47	64.91	0.93	1.66	0.078	3.72	0.341
8	63	F	4.436	129.0	1.10	34.44	30.62	13.29	60.33	13.21	54.67	44.94	0.93	1.65	0.084	3.56	0.518
9	68	M	5.090	128.2	1.12	51.48	33.77	13.17	44.26	13.49	51.83	66.41	0.84	1.62	0.023	3.55	0.277
10	55	F	4.300	129.0	1.10	37.04	12.32	14.13	15.04	14.93	65.14	48.19	1.04	1.71	0.114	3.47	0.270
11	69	M	5.545	131.2	1.08	29.64	9.76	13.23	9.10	15.12	48.16	37.83	0.83	1.74	0.115	3.59	0.263
12	54	M	5.426	131.8	1.09	32.23	17.81	13.35	14.03	12.15	53.73	44.93	0.92	1.65	0.090	3.58	0.185
13	61	M	5.169	128.4	1.11	30.45	15.87	13.08	23.03	13.84	47.86	52.58	0.75	1.61	0.009	3.49	0.315
14	60	M	4.640	129.9	1.10	30.43	13.74	13.45	18.01	13.35	54.90	42.43	0.91	1.66	0.088	3.52	0.158
15	53	M	4.649	130.4	1.10	40.83	28.98	13.36	31.16	12.62	54.44	49.57	0.93	1.65	0.085	3.63	0.311
16	54	M	4.713	129.6	1.10	45.26	29.27	14.00	33.48	14.82	67.16	67.68	1.02	1.65	0.039	3.52	0.194
17	53	M	5.043	130.0	1.10	37.25	39.18	13.24	66.29	13.15	54.90	48.26	0.91	1.64	0.099	3.60	0.252
18	52	M	4.922	129.7	1.10	29.28	10.50	13.21	19.76	13.39	50.96	46.42	0.85	1.63	0.094	3.56	0.200
19	57	F	4.270	130.3	1.10	33.49	14.47	13.22	17.35	13.27	53.18	46.18	0.90	1.64	0.097	3.55	0.228
20	61	F	5.143	129.2	1.10	40.09	15.72	14.09	16.78	14.83	65.17	53.40	1.04	1.70	0.100	3.47	0.265
21	57	F	4.881	129.9	1.10	35.31	14.57	13.26	15.58	13.30	52.07	48.97	0.87	1.63	0.094	3.57	0.442
22	44	F	4.170	131.0	1.10	55.25	35.78	13.23	19.51	12.55	52.36	59.58	0.93	1.61	0.097	3.61	0.335
23	51	M	4.827	130.6	1.08	30.77	12.80	13.60	41.10	14.20	49.79	41.06	0.87	1.90	0.094	3.56	0.337
24	65	M	4.565	129.6	1.10	38.91	25.76	13.20	35.01	13.20	51.94	50.28	0.87	1.62	0.043	3.58	0.212
25	67	F	4.718	129.2	1.10	40.29	31.78	13.75	60.40	14.29	63.50	46.06	1.03	1.72	0.025	3.46	0.412
Mean	59	–	4.799	129.9	1.10	37.74	21.72	13.44	29.62	13.66	55.27	49.82	0.91	1.67	0.078	3.56	0.294
SD	6	–	0.351	0.9	0.01	7.77	10.06	0.35	17.90	0.82	6.15	7.83	0.07	0.06	0.036	0.06	0.089

SD Standard deviation, *F* female, and *M* male

Table 5 Comparisons of model parameters between the treated hypertensive patients ($n = 25$) and the subgroup of normotensive subjects matched for age ($n = 11$)

Model parameters	HT ($n = 25$) [mean ± SD]	NT ($n = 11$) [mean ± SD]	p
R_m (mm)	4.799 ± 0.351	4.612 ± 0.672	0.28
Θ_0 (°)	129.9 ± 0.9	130.0 ± 1.1	0.99
λ (−)	1.099 ± 0.008	1.095 ± 0.010	0.27
c (kPa)	37.74 ± 7.77	31.32 ± 5.14	0.01
c_1 (kPa)	21.72 ± 10.06	18.42 ± 11.37	0.29
c_2 (−)	13.44 ± 0.35	13.50 ± 0.32	0.30
$c_{1\text{circ}}$ (kPa)	29.62 ± 17.90	28.33 ± 20.93	0.57
$c_{2\text{circ}}$ (−)	13.66 ± 0.82	14.01 ± 0.85	0.22
α (°)	55.27 ± 6.15	55.35 ± 5.44	0.41
T_m (kPa)	49.82 ± 7.83	42.47 ± 4.14	0.003
λ_0 (−)	0.91 ± 0.07	0.91 ± 0.09	0.50
λ_m (−)	1.67 ± 0.06	1.72 ± 0.08	0.006
a (kPa)	0.078 ± 0.036	0.094 ± 0.050	0.27
b (−)	3.56 ± 0.06	3.50 ± 0.14	0.13

Significance level of the Wilcoxon rank sum test: $p < 0.05$

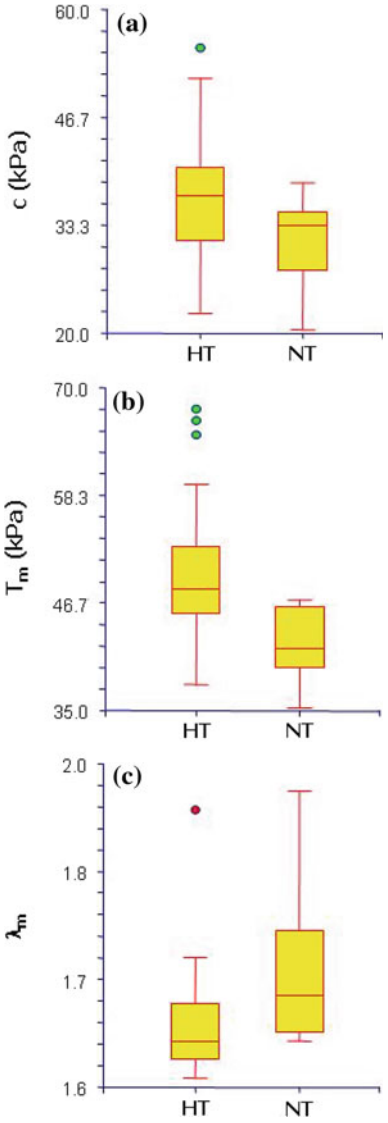


Fig. 6 Box plots for selected model parameters as a function of group: HT for treated hypertensive patients ($n = 25$) and NT for the subgroup of age-matched normotensive subjects ($n = 11$). Results for the parameters **a** c, **b** T_m, and **c** λ_m. Specifically, the *box* represents the middle 50% of the data, with the top and bottom of the *box* denoting the 25th and 75th percentiles and the *horizontal line* drawn through the box at the median (the 50th percentile) value. Adjacent values are displayed as *T-shaped lines* that extend from each end of the box, with the outliers shown as *circles* that are values outside the upper and lower adjacent values

stressed, anisotropic cylindrical tube subjected to changing radial distensions over a cardiac cycle, with a hyperelastic, active-passive, incompressible response. Moreover, the wall was considered to consist of an isotropic matrix dominated by elastin, four predominant families of oriented fibrillar collagen, and circumferentially oriented smooth muscle that can alter its contractile state. Using this approach, we studied two populations: first, normotensive subjects ranging in age from 21 to 64 years to study effects of aging and, second, treated hypertensive patients ranging in age from 44 to 69 years to quantify effects of hypertension despite antihypertensive treatment. The first major result was that aging appeared to affect significantly the residual stress (through the parameters Θ_0 and R_m), the circumferential stiffness due to collagen fibers (through $c_{1\text{circ}}$, $c_{2\text{circ}}$, and α), and the contractile state (through λ_m). Moreover, aging resulted in higher gradients in circumferential stress, higher axial stresses, and reduced ranges of wall stretches (i.e., decreased distensibility). The second major result was that, by comparing results from treated hypertensive and age-matched normotensive patients, we showed that the proposed mechanical modeling allows one to quantify changes in micro-constituent characteristics and wall stresses. In particular, despite effective lowering of blood pressure, the HT group had an increased stiffness of the elastin-dominated matrix (through the parameter c) and a higher level of smooth muscle activation at a lower level of active stretch (through T_m, λ_m); wall stresses were also higher in HT compared to NT subjects. We submit, therefore, that micro-structurally motivated mechanical models can be used to delineate evolving changes in arterial structure and function based on non-invasively collected clinical data, information that is typically available only via invasive (e.g., intravascular ultrasound) methods or histopathologic examination of excised samples.

4.1 Effects of aging

The present micro-structurally motivated model suggested significant changes in CCAs due to aging of normotensive subjects (e.g., via parameters related to residual stress and

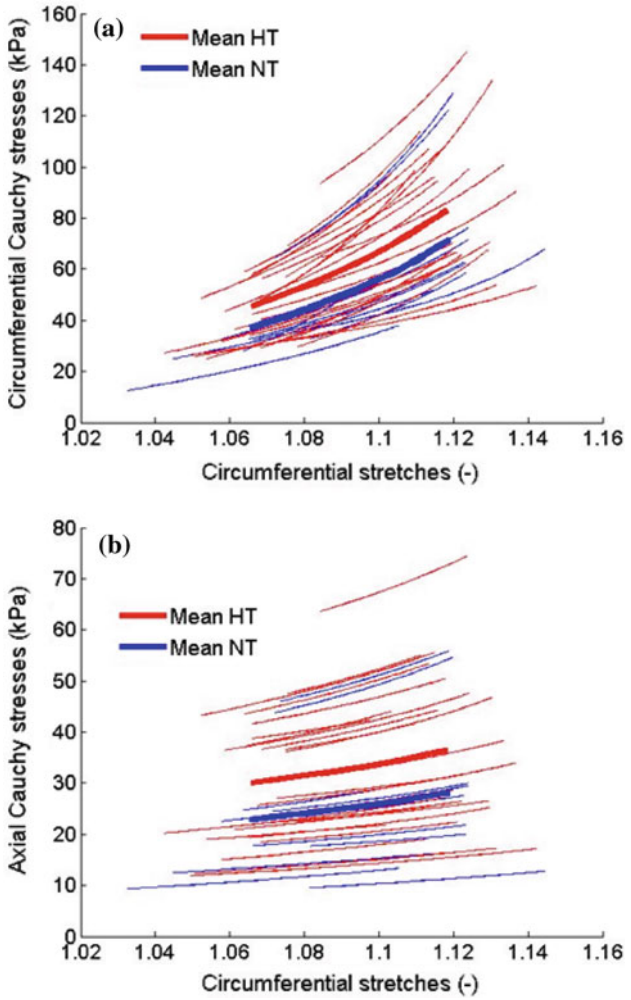


Fig. 7 **a** Circumferential and **b** axial Cauchy stresses as a function of circumferential stretches, expressed at the outer media (r_m), for both the 25 hypertensive patients (HT) and the subgroup of 11 normotensive subjects (NT) matched for age. HT are displayed in *red* and NT in *blue*, with the respective mean values shown by the *darker solid lines*. Mean circumferential and axial stresses (expressed at the mean stretch of the outer medial location) were, respectively, 19 and 32% higher in HT compared to NT subjects

collagen fibers), consistent with previous findings from in vitro studies of arteries excised from animals and humans (cf. Fung 1993; Greenwald 2007; Humphrey 2002). For example, prior reports similarly showed that sector opening, and hence residual stresses, increase in human aortas with aging (Saini et al. 1995). Others (Sommer et al. 2010) reported additional in vitro data (e.g. residual stretches, geometries, distension-extension testing) for eleven human carotid arteries (both common and internal) from subjects aged from 67 to 86 years. They found values of residual stretches that corresponded well to opening angles of 130° published previously for human CCAs (Delfino et al. 1997), consistent with our findings. Although Sommer et al. (2010) did not measure in vivo axial stretches directly, they estimated them from the axial "inversion stretch" of 1.15 ± 0.06 found via in vitro testing, which reflects the in situ axial stretch. We found similar estimated values for in vivo axial stretch despite the different methods of inference.

Unfortunately, detailed histomechanical information on human carotid arteries is scarce for many reasons, including the difficulty of obtaining fresh healthy human tissues for in vitro study and the use of either overly simple or purely phenomenological mechanical models to interpret the limited clinical data (e.g., which lacks direct measurements in vivo of axial forces, perivascular tethering, and residual stresses). Nevertheless, in vivo clinical findings on aging reveal significant decreases in CCA distensibility (Laurent et al. 1994a) and increases in CCA stiffness (Bussy et al. 2000). Similarly, we found that the related stiffness parameters $c_{1\mathrm{circ}}$, $c_{2\mathrm{circ}}$ and c_2 (for collagen fibers) increased with aging. Although these larger values may be interpreted as a stiffening of individual collagen fibers (Wuyts et al. 1995), an increasing of the ratio of collagen to elastin (Sommer et al. 2010), or an increasing of the cross-linking of collagen meshes (Greenwald 2007), we can safely infer that changes in fibrillar collagen contribute significantly to the increased circumferential stiffness in aging. Indeed, the significant increase in the fiber angle α could reflect a reorientation of diagonal collagen fibers toward the circumferential direction with age, which would further contribute to stiffening and limit wall distensibility. Molecular mechanisms underlying the increased stiffening of collagen remain to be determined (e.g., possible role of advanced glycation end-products), but its macroscopic manifestation is consistent with the long-standing observation of arterial stiffening with age, which has a marked impact on the hemodynamics (cf. Greenwald 2007).

Based on a different mechanical model, it has similarly been suggested that alterations in the collagen network are fundamental to explaining aortic stiffening with age (Zulliger and Stergiopulos 2007). It is noteworthy, however, that different types of large arteries appear to respond differently to aging. For example, the abdominal aorta can become susceptible to aneurysm formation in old age while the CCA dilates but almost never becomes aneurysmal. It is well known that elastin degrades or fragments locally in abdominal aortic aneurysms and it is thought that this loss of structural integrity of elastic fibers contributes to the dilatation (Humphrey 2002). Recall, however, that we did not find a significant change in the value of the elastin associated parameter c due to aging (Table 3). This absence of change in c may reflect opposing changes in the quantity (decrease in amount) and properties (increase in stiffness) of elastin in aging CCAs (cf. Greenwald 2007). Increasing calcification of fragmented elastin and neointimal involvement due to aging could also contribute to occlusive disease, which is prevalent in CCAs but not abdominal aortas. Alternatively, because the parameter c embodies both the proportion of the elastin-dominated

matrix and its stiffness, damage to elastin could also be compensated in part by increased production, though it is thought that nearly all of the structurally significant elastin is produced in the perinatal period.

On the other hand, the parameter λ_m, which represents the stretch at which smooth muscle contraction is maximal, increased with aging. This increased value could be explained by several mechanisms; for example, age-associated dilatation may reflect an inability of smooth muscle contractility to counteract increased systolic or pulse pressures that arise during aging (Learoyd and Taylor 1966).

Finally, looking graphically at the transmural stress distributions at given stretches, older subjects had somewhat higher stress than younger ones, although the difference did not reach significance due in part to considerable overlap. This observation may reflect either a different sensitivity of subjects to the aging process or adaptive phenomenon, limiting the increase in stress with stretch (Bussy et al. 2000). Computed levels of wall stresses in the NT population were comparable to those assessed previously for in vivo human abdominal aorta (Stalhand et al. 2004) and of the same order of magnitude as mean values estimated using Laplace's relation. This agreement gives further confidence in the overall model and associated micro-constituent interpretations.

4.2 Effects of hypertension

Results from hypertensive patients were analyzed and compared to those from age-matched normotensive subjects. Although all HT patients had been treated and had well-controlled blood pressures, there were measurable alterations in arterial properties and wall stresses. We previously showed that hypertension associates with CCA dilatation, increased circumferential stress caused by insufficient thickening of the intimal-medial layer, and increased stiffness at operating pressure; one prior major finding, however, was that stiffness was not greater when arteries were studied at identical wall stress (Boutouyrie et al. 1999; Bussy et al. 2000; Laurent et al. 1994a,b). Using the present model, we additionally found significant differences in parameters related to wall micro-constituents, mostly for the elastin-dominated matrix and smooth muscle function. Whereas most prior biomechanical modeling studies relied on animal experiments (e.g., see review by Humphrey 2002), prior clinical studies focused more on functional characteristics, diameters, pulse wave velocities, and derived characteristics that estimate arterial stiffness (e.g. see review by Cheng et al. 2002). Collectively, these results suggest altered structural and functional characteristics associated with hypertension that result from a non-physiologic accumulation of extracellular matrix and dysfunctional endothelial or smooth muscle cells. The modeling results presented herein revealed an unexpected increase in the parameter c (related to the elastin-dominated matrix) in treated HT compared to age-matched NT subjects, with no significant changes in the parameters related to collagen. As mentioned above, increases in the value of c may result from an increased mass fraction, an increased stiffness of the elastin-dominated matrix, or both. Consistent with the discussion related to aging, it may be that despite an expected increased fragmentation or degradation of elastin in HT patients due to increased pulse pressure, there may have also been an increased calcification of the elastin (cf. Greenwald 2007) that resulted in an overall increased stiffening of the elastic matrix. This provocative possibility needs to be explored further, noting that there may be significant differences in CCA changes in treated HT humans versus non-treated humans as well as animals used commonly in hypertension studies.

In addition, our model suggested significant changes in the contractile response in HT patients, as reflected by an increase in the parameter T_m and a decrease in the parameter λ_m (cf. Fig. 6). This finding may suggest that the basal smooth muscle tone was higher at a lower circumferential stretch in HT patients compared to NT subjects. Increased vasoconstriction is thought to be a primary contributor to hypertension. High sympathetic tone is a constant feature of early stages of hypertension, and the most potent antihypertensive drugs, such as ACE (angiotensin-converting enzyme) inhibitors and calcium channel blockers are vasodilating drugs (Mancia et al. 2007). That smooth muscle tone was increased and active stretch at maximum contraction was decreased in well-controlled hypertensive patients (all treated by vasodilating drugs, Table 1) suggests that treatment is unable to fully counteract hypertension associated vasoconstriction.

The proposed model also confirmed that stresses experienced by the CCA wall tend to be higher in hypertensive patients than in normotensive subjects—on average, 1.2-fold higher in circumferential and 1.3-fold higher in axial directions—despite ranges of stretch being similar in the HT and NT subjects. Nevertheless, for a given value of wall stress, circumferential stretches were lower for HT than NT subjects, suggesting the CCAs were stiffer in the HT than the NT subjects consistent with the common clinical observation of decreased distensibility in hypertension.

4.3 Consideration of methods

We have previously discussed in detail our methods for estimating model parameters from non-invasively measured in vivo data (Masson et al. 2008). It is important to note, however, that neither age nor hypertension was introduced at any stage of the optimization process, thus avoiding any a priori biasing of the best-fit values of the model parameters. There is a need, however, to

validate model predictions of wall micro-constituents from histopathological data.

The modeling procedure requires optimization of 14 geometric, structural, or material parameters. Estimation of a large numbers of parameters in a non-linear model by regression is often ill-posed, due in part to the possible existence of multiple local minima (Stalhand et al. 2004). The proposed approach is limited by incomplete in vivo human data and could possibly be improved by a complementary stochastic approach that investigates confidence intervals for estimated best-fit values of model parameters (Ferruzzi et al. 2010). Nevertheless, we employed a previously validated constitutive relation that is motivated by the underlying micro-structure of the wall and physiologic conditions, hence rendering parameters with physical meaning that are all well bounded. In this way, we were able to prescribe reasonable initial guesses for each parameter based on in vitro results in literature. As seen in Fig. 3, estimated parameters yielded pressure–time curves that agreed very well with those measured experimentally. Furthermore, the estimated parameters obtained from the minimization problem were not saddle points as revealed by the perturbation analysis of the solution vectors. Nevertheless, additional methods or studies could improve the parameter identification and should be pursued (cf. Stalhand et al. 2004). These include the following: (1) using a global search algorithm to identify the unique minimum despite possible non-convexity of the objective function, (2) using other objective functions that are ensured to be convex in the physiological region, and/or (3) reducing the number of model parameters by modifying the constitutive relation.

It should also be noted that, although many of the material parameters (c, c_1,…) in the present model derive from micro-structural motivations, they still represent macroscopic characteristics because quantities such as constituent mass fraction, inherent stiffness, and differing times since production are lumped together. Consequently, several phenomena may be competing within a unique parameter (e.g. mass reduction versus stiffening of elastin), possibly resulting in complex interpretations and the need for further investigation. Other constitutive relations having more micro-structurally descriptive parameters should thus be evaluated even though many of the associated values of the parameters could be more difficult to confirm, even in vitro.

Finally, we can suggest possible extensions for similar modeling, noting that this issue was discussed in part previously (Masson et al. 2008). Including residual stresses in modeling is necessary because they strongly influence the stress fields within the wall (Cardamone et al. 2009; Chuong and Fung 1986; Humphrey 2002). Yet, it is not possible to assess the zero-stress state in vivo, and models can be sensitive to the estimated parameters. Moreover, residual stresses appear to be three-dimensional, heterogeneous, and layer-specific (Alastrué et al. 2007; Holzapfel et al. 2007; Sommer et al. 2010). Further improvements are thus needed in modeling and evaluating residual stresses, particularly for human arteries as a function of aging and disease progression. Although few studies account for the effects of perivascular tissue on the wall, this factor likely has a significant impact on in vivo arterial behavior. We chose to model the effects of both the adventitia and perivascular tethering on the intimal-medial layers via a lumped parameter exponential relation that depends on the cardiac cycle as proposed previously (Humphrey and Na 2002). There is a pressing need for increased attention to the importance and modeling of perivascular tissues. Similarly, few models include the effects of smooth muscle contractility even though such contributions to wall mechanics are both fundamental to normal behavior and altered tremendously in most vascular diseases (Humphrey 2002). We used a well-known phenomenological model (Rachev and Hayashi 1999), yet there is a need for further investigation, particularly that addressing better the underlying cytoskeletal structure and biochemistry of smooth muscle contractility (cf. Stalhand et al. 2008). There is similarly a need to quantify better the multiaxial characteristics of smooth muscle contractility and, consequently, to improve modeling of the active contribution by smooth muscle. A circumferential orientation of vascular smooth muscle cells has been consistently observed throughout the human cerebral vasculature (Walmsley et al. 1983), but has not been well studied in the human CCA.

In conclusion, the proposed continuum mechanical framework offers new insight into the mechanics of common carotid arteries in normotensive subjects of different ages and in treated hypertensive patients. The proposed 3D model includes residual stresses, effects of perivascular tissues, and changing loads over a cardiac cycle; it was built by combining a phenomenological description of smooth muscle contractility with a hyperelastic strain–energy function for the passive response, which depended on the behavior of an elastin-dominated isotropic matrix and four predominant families of collagen fibers. Such an approach enables one to infer both macroscopic characteristics such as overall stiffness and micro-structural information on possible changes in the contributions by individual constituents. This increased capability offers particular promise for clinical investigations based on in vivo data, such as the non-invasive study of the effects of aging and hypertension on arterial stiffness and stress fields within the wall, wherein micro-structural information is limited but potentially very important nonetheless.

Conflict of interest The authors declare no conflict of interest.

References

Alastrué V, Pena E, Martinez MA, Doblaré M (2007) Assessing the use of the "opening angle method" to enforce residual stresses in patient-specific arteries. Ann Biomed Eng 35(10):1821–1837

Baek S, Gleason RL, Rajagopal KR, Humphrey JD (2007) Theory of small on large: potential utility in computations of fluid-solid interactions in arteries. Comp Methods Appl Mech Eng 196(31–32):3070–3078

Beaussier H, Masson I, Collin C, Bozec E, Laloux B, Calvet D, Zidi M, Boutouyrie P, Laurent S (2008) Carotid plaque, arterial stiffness gradient, and remodeling in hypertension. Hypertension 52(4):729–736

Boutouyrie P, Bussy C, Lacolley P, Girerd X, Laloux B, Laurent S (1999) Association between local pulse pressure, mean blood pressure, and large-artery remodeling. Circulation 100(13):1387–1393

Bussy C, Boutouyrie P, Lacolley P, Challande P, Laurent S (2000) Intrinsic stiffness of the carotid arterial wall material in essential hypertensives. Hypertension 35(5):1049–1054

Cardamone L, Valentin A, Eberth JF, Humphrey JD (2009) Origin of axial prestretch and residual stress in arteries. Biomech Model Mechanobiol 8(6):431–446

Cheng K-S, Baker CR, Hamilton G, Hoeks APG, Seifalian AM (2002) Arterial elastic properties and cardiovascular risk/event. Eur J Vasc Endovasc Surg 24(5):383–397

Chuong CJ, Fung YC (1983) Three-dimensional stress distribution in arteries. J Biomech Eng 105(3):268–274

Chuong CJ, Fung YC (1986) On residual stresses in arteries. J Biomech Eng 108(2):189–192

Delfino A, Stergiopulos N, Moore JE Jr, Meister JJ (1997) Residual strain effects on the stress field in a thick wall finite element model of the human carotid bifurcation. J Biomech 30(8):777–786

Eddhahak-Ouni A, Masson I, Allaire E, Zidi M (2009) Stochastic approach to estimate the arterial pressure. Eur J Mech A/Solid 28(4):712–719

Ferruzzi J, Vorp DA, Humphrey JD (2010) On constitutive descriptors of the biaxial mechanical behaviour of human abdominal aorta and aneurysms. J R Soc Interface (in press). doi:10.1098/rsif.2010.0299

Fridez P, Rachev A, Meister JJ, Hayashi K, Stergiopulos N (2001) Model of geometrical and smooth muscle tone adaptation of carotid artery subject to step change in pressure. Am J Physiol Heart Circ Physiol 280(6):H2752–H2760

Fung YC (1993) Biomechanics: mechanical properties of living tissues. Springer, New York

Gasser TC, Holzapfel GA (2002) A rate-independent elastoplastic constitutive model for biological fiber-reinforced composites at finite strains: Continuum basis, algorithmic formulation and finite element implementation. Comput Mech 29(4–5):340–360

Gasser TC, Ogden RW, Holzapfel GA (2006) Hyperelastic modelling of arterial layers with distributed collagen fibre orientations. J R Soc Interface 3(6):15–35

Greenwald SE (2007) Ageing of the conduit arteries. J Pathol 211(2):157–172

Hariton I, deBotton G, Gasser TC, Holzapfel GA (2007) Stress-modulated collagen fiber remodeling in a human carotid bifurcation. J Theor Biol 248(3):460–470

Holzapfel GA, Gasser TC, Ogden RW (2000) A new constitutive framework for arterial wall mechanics and a comparative study of material models. J Elast 61(1–3):1–48

Holzapfel GA, Sommer G, Auer M, Regitnig P, Ogden RW (2007) Layer-specific 3d residual deformations of human aortas with non-atherosclerotic intimal thickening. Ann Biomed Eng 35(4):530–545

Holzapfel GA, Sommer G, Gasser CT, Regitnig P (2005) Determination of layer-specific mechanical properties of human coronary arteries with nonatherosclerotic intimal thickening and related constitutive modeling. Am J Physiol Heart Circ Physiol 289(5):H2048–H2058

Humphrey JD (2002) Cardiovascular solid mechanics: cells, tissues, and organs. Springer, New York

Humphrey JD, Eberth JF, Dye WW, Gleason RL (2009) Fundamental role of axial stress in compensatory adaptations by arteries. J Biomech 42(1):1–8

Humphrey JD, Na S (2002) Elastodynamics and arterial wall stress. Ann Biomed Eng 30(4):509–523

Kelly R, Hayward C, Avolio A, O'Rourke M (1989) Noninvasive determination of age-related changes in the human arterial pulse. Circulation 80(6):1652–1659

Laurent S, Caviezel B, Beck L, Girerd X, Billaud E, Boutouyrie P, Hoeks A, Safar M (1994a) Carotid artery distensibility and distending pressure in hypertensive humans. Hypertension 23(6):878–883

Laurent S, Girerd X, Mourad JJ, Lacolley P, Beck L, Boutouyrie P, Mignot JP, Safar M (1994b) Elastic-modulus of the radial artery wall material is not increased in patients with essential-hypertension. Arterioscler Thromb Vasc Biol 14(7):1223–1231

Learoyd BM, Taylor MG (1966) Alterations with age in the viscoelastic properties of human arterial walls. Circ Res 18(3):278–292

Mancia G, De Backer G, Dominiczak A, Cifkova R, Fagard R, Germano G, Grassi G, Heagerty AM, Kjeldsen SE, Laurent S, Narkiewicz K, Ruilope L, Rynkiewicz A, Schmieder RE, Boudier H, Zanchetti A (2007) 2007 guidelines for the management of arterial hypertension. J Hypertens 25(6):1105–1187

Masson I, Boutouyrie P, Laurent S, Humphrey JD, Zidi M (2008) Characterization of arterial wall mechanical behavior and stresses from human clinical data. J Biomech 41(12):2618–2627

Meinders JM, Brands PJ, Willigers JM, Kornet L, Hoeks AP (2001) Assessment of the spatial homogeneity of artery dimension parameters with high frame rate 2-d b-mode. Ultrasound Med Biol 27(6):785–794

Paini A, Boutouyrie P, Calvet D, Zidi M, Agabiti-Rosei E, Laurent S (2007) Multiaxial mechanical characteristics of carotid plaque: analysis by multiarray echotracking system. Stroke 38(1):117–123

Rachev A, Hayashi K (1999) Theoretical study of the effects of vascular smooth muscle contraction on strain and stress distributions in arteries. Ann Biomed Eng 27(4):459–468

Saini A, Berry C, Greenwald S (1995) Effect of age and sex on residual stress in the aorta. J Vasc Res 32(6):398–405

Schulze-Bauer CA, Holzapfel GA (2003) Determination of constitutive equations for human arteries from clinical data. J Biomech 36(2):165–169

Sommer G, Regitnig P, Koltringer L, Holzapfel GA (2010) Biaxial mechanical properties of intact and layer-dissected human carotid arteries at physiological and supraphysiological loadings. Am J Physiol Heart Circ Physiol 298(3):H898–H912. doi:10.1152/ajpheart.00378.2009

Stalhand J (2009) Determination of human arterial wall parameters from clinical data. Biomech Model Mechanobiol 8(2):141–148

Stalhand J, Klarbring A, Holzapfel GA (2008) Smooth muscle contraction: mechanochemical formulation for homogeneous finite strains. Prog Biophys Mol Biol 96(1–3):465–481

Stalhand J, Klarbring A, Karlsson M (2004) Towards in vivo aorta material identification and stress estimation. Biomech Model Mechanobiol 2(3):169–186

van Loon P (1977) Length-force and volume-pressure relationships in arteries. Biorheology 14(4):181–201

Walmsley J, Campling M, Chertkow H (1983) Interrelationships among wall structure, smooth muscle orientation, and contraction in human major cerebral arteries. Stroke 14(5):781–790

Wuyts FL, Vanhuyse VJ, Langewouters GJ, Decraemer WF, Raman ER, Buyle S (1995) Elastic properties of human aortas in relation to age and atherosclerosis: a structural model. Phys Med Biol 40(10):1577–1597

Zulliger MA, Fridez P, Hayashi K, Stergiopulos N (2004) A strain energy function for arteries accounting for wall composition and structure. J Biomech 37(7):989–1000

Zulliger MA, Stergiopulos N (2007) Structural strain energy function applied to the ageing of the human aorta. J Biomech 40(14):3061–3069